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Loneliness, Identity, and Mental Health Among Emerging Adults

Emily Garrard[a,]

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Loneliness among college-aged adults has reached crisis levels, yet its relationship to mental health and most effective interventions remain poorly understood. Using data from over 60,000 emerging adults (ages 18-24) drawn from the Healthy Minds Study 2024-2025 data, we examine how loneliness varies by identity sexual orientation, gender, and race, and whether campus belonging and perceived stigma moderate its effects on depression and anxiety. Subgroup analyses reveal that sexual minority and gender minority students report significantly higher loneliness than heterosexual and cisgender peers. Multivariate OLS regression confirms that loneliness independently predicts both depression (PHQ-9) and anxiety (GAD-7) after controlling for identity, belonging, stigma, and age. Perceived stigma significantly amplifies the effect of loneliness on depression, while campus belonging independently reduces depression and anxiety but does not significantly moderate the loneliness–depression pathway. These findings highlight identity-specific vulnerability to loneliness and point to stigma reduction and belonging as promising targets for campus mental health intervention.

loneliness | mental health | emerging adults | minority stress | belonging

Introduction

Loneliness among college-aged adults has reached crisis levels. The U.S. Surgeon General's 2023 advisory identifies loneliness as a public health epidemic, with prevalence highest among adults aged 18–24 (1). Yet most research treats loneliness either as a purely individual experience–a symptom of depression or social anxiety–or as a broad societal phenomenon driven by structural disconnection. This framing misses something important: loneliness and mental illness are deeply intertwined, each reinforcing the other in ways that make both harder to address in isolation. Cacioppo et al. (2006) demonstrate that loneliness predicts depressive symptoms even after controlling for perceived stress and social support, and Moeller and Seehuus (2019) find that this loneliness-to-depression pathway is particularly strong in college samples (2, 3). Hawkley and Cacioppo (2010) further show that loneliness produces hypervigilance for social threat–a cognitive pattern that causes lonely individuals to misread others' intentions and withdraw further, creating a self-reinforcing cycle (4).

Less understood is how this cycle varies across identity groups and campus environments. Meyer's (2003) minority stress theory predicts that sexual minority individuals face identity-specific stressors–discrimination, stigma, concealment–that elevate both loneliness and depression risk above and beyond general life stress (5). Vasileiou et al. (2021) extend this to college populations, noting that marginalized students face compounding vulnerabilities that campus wide interventions rarely address (6). Campus belonging has been proposed as a protective factor, yet its precise role in buffering the loneliness-mental health relationship remains unclear. If stigma around mental health discourages students from seeking connection, the problem may be less about belonging to large campus communities and more about whether any single person notices them.

This paper uses data from over 60,000 emerging adults (ages 18-24) from the Healthy Minds Dataset to examine three questions. First, how does loneliness relate to depression and anxiety in this population? Second, do these relationships differ by sexual orientation, gender, or race? Third, do

Significance

Nearly half of college students meet the clinical threshold for loneliness, yet campus mental health efforts rarely target loneliness directly. We show that sexual minority and gender minority students are disproportionately affected, and that loneliness is the single strongest predictor of both depression and anxiety in a sample of over 60,000 emerging adults–stronger than identity, belonging, or stigma alone. Critically, perceived stigma amplifies how much loneliness hurts, while broad belonging initiatives reduce depression generally but do not protect the loneliest students specifically. These findings suggest that the most effective campus interventions may need to target stigma reduction and identity-specific support rather than belonging broadly defined.

Author affiliations: [a]Dartmouth College

campus belonging and perceived stigma act as protective or risk factors for the loneliness-mental health relationship? We find that sexual minority and gender minority students report significantly higher loneliness than their peers, that loneliness strongly and independently predicts both depression and anxiety, and that perceived stigma–but not campus belonging broadly–significantly amplifies the effect of loneliness on depression. These findings suggest that reducing stigma and fostering one-on-one connection may matter more than large-scale belonging initiatives for the students most at risk.

Data

Source. Data comes from the Healthy Minds Study 2024-2025, a large-scale annual survey of college students conducted during the 2024–25 academic year. The full dataset contains 84,735 respondents across 135 institutions. This analysis restricts the sample to emerging adults aged 18-24, yielding an analytic sample of 17,120 respondents after dropping cases with missing values on key regression variables.

Unit of Analysis. The unit of analysis is the individual student. Each row in the dataset represents one survey respondent's self-reported responses at a single point in time. The survey is cross-sectional, meaning we observe each student once and cannot track changes over time.

Key Variables. *Depression* is measured using the PHQ-9, a nine-item validated screener in which respondents rate the frequency of depressive symptoms over the past two weeks on a 0-3 scale, producing a total score ranging from 0-27. Scores of 10 or above indicate likely clinical depression (7).

Anxiety is measured using the GAD-7, a seven-item validated screener rated on the same 0-3 scale, producing a total score of 0-21. Scores of 10 or above indicate likely clinical anxiety (8).

Loneliness is measured using the UCLA-3 Loneliness Scale, a three-item measure rated on a 1-3 scale producing a total score of 3-9. Scores of 6 or above are used as a conventional loneliness threshold (9).

Campus belonging is a composite score derived from 3 survey items measuring students' sense of connection to their campus community. Higher scores indicate greater belonging.

Perceived stigma measures students' beliefs about how negatively their campus environment views mental health struggles. Higher scores indicate greater perceived stigma.

Identity variables include gender (Woman, Man, Nonbinary, Genderqueer, Trans, Self-described), sexual orientation (Heterosexual, Lesbian, Gay, Bisexual, Pansexual, Queer, Questioning, Asexual, Self-described), and race (White, Racial minority).

Limitations. Several limitations deserve note. First, the data are cross-sectional, meaning we cannot establish causal direction between loneliness and mental health outcomes. As Cacioppo et al. (2006) note, loneliness and depression are likely reciprocally reinforcing over time (2). Second, the survey relies on self-report, which may introduce social desirability bias, particularly for stigmatized experiences like loneliness and mental illness. Third, the sample is restricted to students at participating institutions and may not generalize to non-college emerging adults or students at non-participating schools. Fourth, 44,168 respondents were dropped from regression analyses due to missing values on one or more variables, which may introduce bias if missingness is not random.

Sample Characteristics. Table 1 summarizes the demographic composition of the analytic sample.

Methods

Data Cleaning. Raw survey data were cleaned in `00_clean_data.ipynb`. Composite scores for depression (PHQ-9), anxiety (GAD-7), and loneliness (UCLA-3) were computed by summing item-level responses and converting from the survey's 1-4 coding back to the standard 0-3 scale. Campus belonging and

Table 1. Sample characteristics of emerging adults aged 18–24 (HMS 2024–25, n = 61,288).

	n	%
Gender		
Woman	39,186	63.9
Man	16,227	26.5
Nonbinary	2,325	3.8
Genderqueer	1,254	2.0
Trans	1,131	1.8
Self-described	126	0.2
Sexual Orientation		
Heterosexual	37,593	61.3
Bisexual	9,446	15.4
Pansexual	1,780	2.9
Queer	2,165	3.5
Lesbian	2,283	3.7
Gay	1,247	2.0
Questioning	1,240	2.0
Asexual	1,273	2.1
Self-described	732	1.2
Race		
White	30,381	49.6
Racial minority	29,891	48.8
Missing/Other		1.6
Age	Mean	SD
	20.2	1.7

perceived stigma composites were constructed with reverse-coding. Gender, sexual orientation, and race were recoded from raw survey items into categorical variables with clearly defined reference groups (women, heterosexual, and white, respectively). The sample was then restricted to respondents aged 18-24, yielding an analytic sample of 61,288 emerging adults. The cleaned dataset was saved to `data/emerging_adults_clean.csv` and loaded by all subsequent notebooks.

No merging of separate data sources was required; all variables used in this analysis were drawn from a single HMS survey file (`HMS_2024-2025_PUBLIC_instchars.csv`). The unit of analysis in both the raw data and the analytic sample is the individual student respondent.

Variables. Outcome variables are PHQ-9 total score (depression), GAD-7 total score (anxiety), and UCLA-3 total score (loneliness), each treated as continuous. Clinical thresholds used for descriptive purposes are PHQ-9 $\geq$ 10 (likely depression), GAD-7 $\geq$ 10 (likely anxiety), and UCLA-3 $\geq$ 6 (loneliness). The primary predictor of interest is UCLA-3 loneliness score. Campus belonging and perceived stigma are continuous moderators constructed from multi-item HMS scales; higher scores indicate greater belonging and greater perceived stigma. Dummy-encoded covariates include gender (reference: women), sexual orientation (reference: heterosexual), and race (reference: white). Age is included as a continuous covariate. All variable construction is documented in `00_clean_data.ipynb`.

Descriptive and Subgroup Analysis. Descriptive analyses were conducted in `01_explore_data.ipynb`. Means, standard deviations, and clinical screening rates were computed for each outcome. Pairwise Pearson correlations were computed among all three outcomes. To examine identity-based disparities, mean loneliness scores were compared across binary sexual orientation (heterosexual vs. sexual minority) and race (white vs. racial minority) groups using Welch's independent-samples t-tests, which do not assume equal variances and are preferred when group sizes differ substantially. Effect sizes were reported as Cohen's d, the standardized mean difference using a pooled standard deviation, interpreted using conventional benchmarks of small (0.2), medium (0.5), and large (0.8). With a sample of over 60,000 students, nearly any difference will reach statistical significance; Cohen's d is therefore important for practical importance. Students were also divided into tertiles of campus belonging and perceived stigma to illustrate relationships with loneliness.

Regression Models. Four multivariate OLS regression models were estimated in `02_analysis.ipynb` using `statsmodels`, which provides coefficient estimates alongside standard errors, t-statistics, p-values, and 95% confidence intervals. All models included UCLA-3 loneliness, campus belonging, self-stigma, perceived stigma, and age as continuous predictors, along with dummy-encoded gender, sexual orientation, and race. Regression analyses used complete cases only; 44,168 respondents with missing values on any model variable were excluded, leaving a regression sample of 17,120. Descriptive analyses used all available data for each variable.

The four models are: (1) *Depression model* – OLS predicting PHQ-9 total score from all predictors; (2) *Anxiety model* – OLS predicting GAD-7 total score; (3) *Loneliness model* – OLS predicting UCLA-3 score from identity, belonging, stigma, and age only, to test whether identity predicts loneliness independently of campus environment; (4) *Interaction models* – two additional depression models each adding a single interaction term (loneliness × perceived stigma; loneliness × belonging) to test whether stigma amplifies and belonging buffers the loneliness-depression relationship.

Missing Data. Missing data was handled differently across the stages of analysis. Descriptive analyses and subgroup comparisons used all available data for each variable, yielding slightly different n values across measures (PHQ-9: $n = 52{,}314$; GAD-7: $n = 51{,}952$; UCLA-3: $n = 52{,}717$). Regression analyses required complete cases on all model variables simultaneously; 44,168 respondents were excluded on this basis, leaving a regression sample of 17,120. Missingness was not examined as a function of key predictors, so it is possible that the regression sample differs systematically from the full analytic sample–for example, if students with more severe mental health symptoms were less likely to complete all survey items. This is a limitation of the complete-case approach and should be noted when interpreting regression results.

Software and Analytical Decisions. All analyses were conducted in Python 3. Data cleaning used `pandas`; descriptive statistics and correlations used `scipy.stats`; regression models were estimated using `statsmodels.api`, which produces full model summaries including R^2, coefficient estimates, standard errors, t-statistics, p-values, and 95% confidence intervals. OLS was chosen because all outcomes (PHQ-9, GAD-7, UCLA-3) are treated as continuous. Figures were produced using `matplotlib` and `seaborn` and saved as PNG files to the `output/` directory.

Although we use regression models, this analysis is correlational rather than causal. Because the data was collected at a single point in time, we cannot determine the direction of the relationship–for example, loneliness may cause depression, but depression may equally cause loneliness. We also cannot rule out the influence of unmeasured variables, such as pre-existing mental health history, that could explain the associations we observe. The goal of this analysis is not to establish causation but to describe how strongly loneliness and mental health are linked and identify which students are most at risk. Establishing a causal relationship would require data collected over time, a natural experiment, or a randomized study.

Results

Sample Characteristics. Table 2 presents descriptive statistics for the full analytic sample of 61,288 emerging adults aged 18-24. Clinical screening rates were substantial: 31.6% met the PHQ-9 ≥ 10 threshold for likely depression, 28.8% met the GAD-7 ≥ 10 threshold for likely anxiety, and 44.2% scored at or above the UCLA-3 ≥ 6 loneliness threshold. All three outcomes were moderately byt significantly correlated: depression-anxiety $r = 0.741$, depression-loneliness $r = 0.541$, anxiety-loneliness $r = 0.473$.

Figure 1 shows that mean PHQ-9 and GAD-7 scores increase monotonically across loneliness levels, crossing the clinical threshold in the "quite lonely" range.

Table 2. Descriptive statistics for the analytic sample of emerging adults aged 18-24 (HMS 2024-25). Clinical screening rates are based on standard cutoffs: PHQ-9 $\geq$ 10, GAD-7 $\geq$ 10, UCLA-3 $\geq$ 6. The regression sample (n = 17,120) excludes respondents with missing values on any model variable.

Variable	n	Mean (SD)	Range	% at/above threshold
Depression (PHQ-9)	52,314	8.40 (6.22)	0-27	31.6%
Anxiety (GAD-7)	51,952	7.61 (5.78)	0-21	28.8%
Loneliness (UCLA-3)	52,717	5.54 (1.94)	3-9	44.2%

Full analytic sample: n = 61,288. Regression sample (complete cases): n = 17,120.

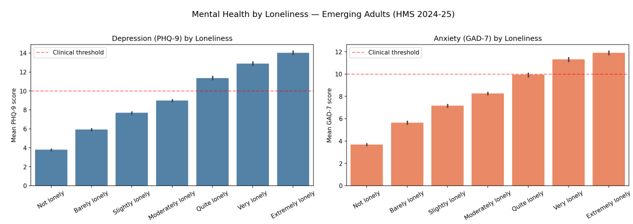

Fig. 1. Mean PHQ-9 depression and GAD-7 anxiety scores across UCLA-3 loneliness levels among emerging adults aged 18-24 (HMS 2024-25). Scores increase monotonically with loneliness across both outcomes. Red dashed line indicates the clinical threshold ($\geq$ 10). Error bars represent 95% confidence intervals.

Subgroup Differences in Loneliness. Sexual minority students reported significantly higher loneliness than heterosexual students (M $= 6.12$ vs. M $= 5.24$, Cohen's $d = 0.468$), a near-medium effect. Questioning and pansexual students showed the highest loneliness within this group, suggesting that identity uncertainty and social invisibility compound the stressors of minority status. Figure 2 illustrates this gap; sexual minority students cross the clinical loneliness threshold on average while heterosexual students do not. By contrast, racial minority students showed a statistically significant but negligible raw difference from white students ($d = 0.050$), and racial minority status was not a significant predictor of loneliness in the multivariate model ($p = 0.074$), indicating the raw difference is explained by other factors once identity and campus variables are controlled.

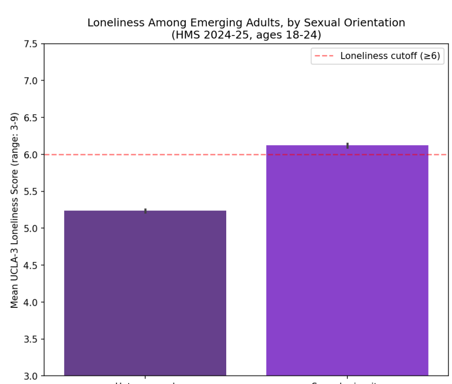

Fig. 2. Mean UCLA-3 loneliness scores by sexual orientation among emerging adults aged 18-24 (HMS 2024-25). Sexual minority students average 6.12 vs. 5.24 for heterosexual peers, crossing the loneliness threshold ($\geq$ 6). Cohen's d = 0.468. Error bars represent 95% confidence intervals.

Multivariate Regression Results. In the depression model ($R^2 = 0.360$), loneliness was the strongest predictor ($\beta = 1.259$, $p < .001$): each one-point increase in UCLA-3 score was associated with a 1.26-point increase in PHQ-9. Campus belonging was a modest independent protective factor ($\beta = -0.262$, $p < .001$). Both self-stigma ($\beta = 0.164$) and perceived stigma ($\beta = 0.081$) were significant but small relative to loneliness. Sexual minority identity was associated with substantially higher depression, with pansexual ($\beta = 2.512$) and queer students ($\beta = 1.968$) showing the largest gaps. Gender minority students also reported higher depression, while men reported lower depression than women ($\beta = -0.970$).

The anxiety model ($R^2 = 0.285$) showed the same broad pattern: loneliness remained the strongest predictor ($\beta = 1.088$, $p < .001$) and belonging was again modestly protective ($\beta = -0.163$). Men showed a substantially larger anxiety gap than in the depression model ($\beta = -2.149$), and several gender minority groups significant for depression did not reach significance for anxiety, suggesting gender minority identity may be more specifically tied to depression than anxiety.

Interaction Models: Stigma Amplifies; Belonging Does Not Buffer. The stigma interaction model ($R^2 = 0.361$) revealed a significant positive interaction between loneliness and perceived stigma ($\beta = 0.029$, $p < .001$): students in higher-stigma environments experience a steeper increase in depression per unit of loneliness. Although the coefficient is small in absolute terms, it is reliable and cumulative across the range of the stigma scale. Perceived stigma alone was not a significant independent predictor in this model ($p = 0.038$), suggesting its primary role is as a moderator amplifying the harm of loneliness rather than a direct driver of depression.

The belonging interaction model ($R^2 = 0.361$) produced a negative but non-significant interaction term ($\beta = -0.009$, $p = 0.053$). Belonging and loneliness have mostly independent additive effects on depression: belonging shifts depression scores down uniformly across all loneliness levels but does not meaningfully change the slope of the loneliness-depression relationship. Students who are lonely with high belonging are not substantially more protected from depression than students who are lonely with low belonging. Broad belonging initiatives may therefore reduce depression generally but are unlikely to specifically interrupt the loneliness-depression pathway for the most vulnerable students.

Discussion

This study finds that loneliness is the strongest and most consistent predictor of both depression and anxiety among emerging adults, independent of identity, belonging, and stigma. Sexual minority and gender minority students report substantially higher loneliness than their peers even after controlling for campus factors, with questioning and pansexual students showing the largest gaps–suggesting that identity uncertainty and social illegibility compound loneliness beyond minority status alone. Perceived stigma significantly amplifies the effect of loneliness on depression, while campus belonging reduces depression and anxiety independently but does not specifically buffer the loneliness-depression pathway. This distinction has practical implications: large-scale campus belonging initiatives may raise average well-being but are unlikely to reach the most vulnerable students. Interventions targeting one-on-one connection and stigma reduction may be more effective for students at the intersection of loneliness and minority identity.

Several limitations deserve note. The data are cross-sectional, preventing causal inference. The regression sample of 17,120 is substantially smaller than the full sample of 61,288 due to missing data, which may introduce bias. Future research should examine whether close relationship quality–rather than campus belonging broadly–more strongly moderates the loneliness-depression relationship, and should use longitudinal data to establish the direction of effects between loneliness and mental health over time.

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Appendix: Agentic Analysis

Overview. This project used Claude as an AI coding assistant throughout my analysis. Here is a critical reflection on what was asked, what was accepted, what was rejected or modified, and where the assistant made errors.

What I Asked the Assistant. First, I asked Claude to help me find publicly downloadable mental health datasets for college students, which led to the Healthy

Minds Study (HMS) 2024-25. Claude helped with data loading, sorting, and interpreting data (reading the CSV, filtering columns, computing PHQ-9 and GAD-7 totals, building loneliness and stigma composite variables, and creating clinical screening flags). It also helped with statistical analysis (Welch's t-tests, multivariate OLS regression, and interaction models), and interpretation of results. I also asked for help with code formatting, GitHub repository structure, and paper writing, including the introduction, framing of findings, and how to defend results with actual numbers during the presentation.

What I Accepted. The overall analytical structure–Welch's t-tests for raw group comparisons, OLS regression for controlled effects, and Cohen's d for effect sizes–was Claude's recommendation. I accepted the `min_count=9` argument in the PHQ-9 sum, which protects against artificially low scores when respondents skip items. I also accepted Claude's framing for interpreting the three stigma scales (personal, perceived, and self-stigma). Claude's suggestion to use bootstrap confidence intervals to characterize uncertainty was accepted and helped shape how I described statistical results in the paper.

What I Rejected or Modified. **Cross-dataset comparison.** Claude initially supported exploring a comparison between college and non-college loneliness using BRFSS alongside Healthy Minds Study. I ultimately dropped this direction and all analyses were HMS.

Combining ten years of data. Claude walked me through code for concatenating the 2015-2025 HMS datasets. I chose to work with a single year (2024-25) because my research question was about current students, not longitudinal trends.

Where the Assistant Went Wrong. **Stigma scale direction (most significant error).** Claude initially assumed that higher scores on the HMS stigma items (`stig_per_*`, `stig_pcv_*`) indicated more stigma. The resulting correlations were negative–stigma negatively predicting loneliness–which contradicts theory. Claude flagged the anomaly but directed me to the codebook rather than catching the error independently. After I reviewed the codebook, Claude proposed a fix that reverse-coded one item (`stig_pcv_2`), which was also incorrect. A second round of codebook review was required before the correct recoding procedure was established. The corrected composites (all correlations positive and consistent with the literature) were ultimately valid, but required two rounds of correction and my own judgment to recognize that the initial outputs were theoretically implausible.

Guessed variable names. When first constructing the working DataFrame, Claude guessed variable names such as `loneliness` and `phq9_total` that did not match the actual HMS column names (`lonesc` and a computed total I built manually). This produced `KeyError`s that required debugging. Claude's strategy of using keyword searches over `df.columns` was useful after this.

`sklearn` for research regression. Claude's first regression implementation used `sklearn.linear_model.LinearRegression`, which does not output standard errors, confidence intervals, or p-values. I ended up changing it to `statsmodels`.

Reflection. Claude was most helpful helping me write and troubleshoot my code. It was very valuable telling me which statistical choices to make and interpreting my results. The most important skill I learned through working with Claude was treating Claude's code as a starting hypothesis rather than a final answer–running it, checking outputs against theory, and pushing back when results did not make sense.